Aaryan Degama

aaryandegama@gmail.com | +91-8320894345 | Surat, India | Portfolio | GitHub | LinkedIn | Codeforces (1446)

Summary

Third-year B.Tech (IT – Business Informatics) student at IIIT Allahabad focused on systems programming, computer vision, and ML systems. Currently building Vanta, a privacy-first Android app that finds photos and files entirely on-device using CLIP, InsightFace (`buffalo_l`), BERT, and ONNX Runtime, all powered by a portable native engine (`libvanta`) built for Android/iOS. Interested in applied ML research, competitive programming, and engineering efficient systems that work at the edge.

Education

Indian Institute of Information Technology, Allahabad (IIIT Prayagraj) Expected May 2028
B.Tech – Information Technology (Business Informatics) | *CGPA: 7.35*

Indian School Sohar, Oman 2022 – 2024
CBSE Class XII – 86% (Apr 2024) | *CBSE Class X – 89% (Apr 2022)*

Technical Skills

Languages: C++, C, Python, JavaScript, Rust (learning)
ML / DL: PyTorch, ONNX Runtime, Transformers; CLIP, InsightFace (`buffalo_sc`, `buffalo_l`), BERT – training, fine-tuning & ONNX export
Systems: Linux (Fedora/Wayland), Git, CMake, JNI, Docker
Cloud: AWS (Amplify Gen 2, Lambda, DynamoDB, AppSync, Cognito, S3), Cedar
Databases: SQLite, sqlite-vec, MongoDB, MySQL
Web / APIs: Node.js, Express, FastAPI, React, TypeScript, REST
Other: Information Retrieval (RAG / Graph RAG), competitive programming

Projects

Vanta – On-Device Semantic Search Engine July 2026 – Present
Android | C++ | ONNX Runtime | SQLite | React Native

- Built a fully on-device semantic photo & file search app for Android – no cloud, no data leaves the device.
- Engineered the retrieval architecture combining CLIP ViT-B/32 (visual embeddings), InsightFace `buffalo_l` (face identity), and BERT-tiny (NER) via ONNX Runtime, with `sqlite-vec` for vector search and a SQLite-backed property graph (entity/relation adjacency model) for people and relationship queries.
- Designed a shared C++ core (`libvanta`) via JNI for Android/iOS portability; implemented a query pipeline (SymSpell correction → NER → entity resolution → CLIP re-ranking) with combined graph + visual-similarity scoring for mixed people/context queries.
- Fine-tuned BERT-tiny on a 5,600-sample IOB2 NER dataset (Gemini-generated) with hard vocabulary splits, and exported to ONNX (opset 18) for on-device Android inference.
- Achieved an average end-to-end search latency of ∼280ms across all query types (short/long, entity/relation, spell-corrected), with rare peaks of ∼430–450ms.

Slate – Cross-Section Timetable & Scheduling Platform Sept 2026
React | TypeScript | AWS Amplify Gen 2 | DynamoDB | Cedar | Team of 3

- Built and shipped a serverless scheduling app (Cognito, AppSync, DynamoDB, Lambda, S3) in a 3.5-day AWS hackathon (First Commit, WeMakeDevs × AWS), gated to institute accounts via a domain-checking pre-sign-up Lambda.
- Wrote the slot-finder as real interval-intersection scheduling across multiple sections/instructors: ranks free-time candidates with explainable scoring and, when no slot works, names the specific section blocking it.
- Authorized every write with Cedar policies evaluated in-Lambda via `cedar-wasm`, logging each allow/deny decision as structured JSON for audit.
- Built the timetable-ingestion pipeline (spreadsheet parsing, cell-format normalization, hour/course validation against each course's own credit legend) with a diff-and-confirm review step before any import.

Achievements & Open Source

- Ranked 30/150 in Terranent50 TensorFlow open-source competition.
- Ranked 16/350 in IIIT Allahabad intra-college open-source competition.
- Codeforces rating 1446 – consistent competitive programmer.
- Top 5 in a 45-minute on-campus AI build sprint by Sarvam AI (Oct 2026), built with Claude Code.

Languages

Gujarati (Native) | English (Fluent) | Hindi (Professional Working Proficiency)